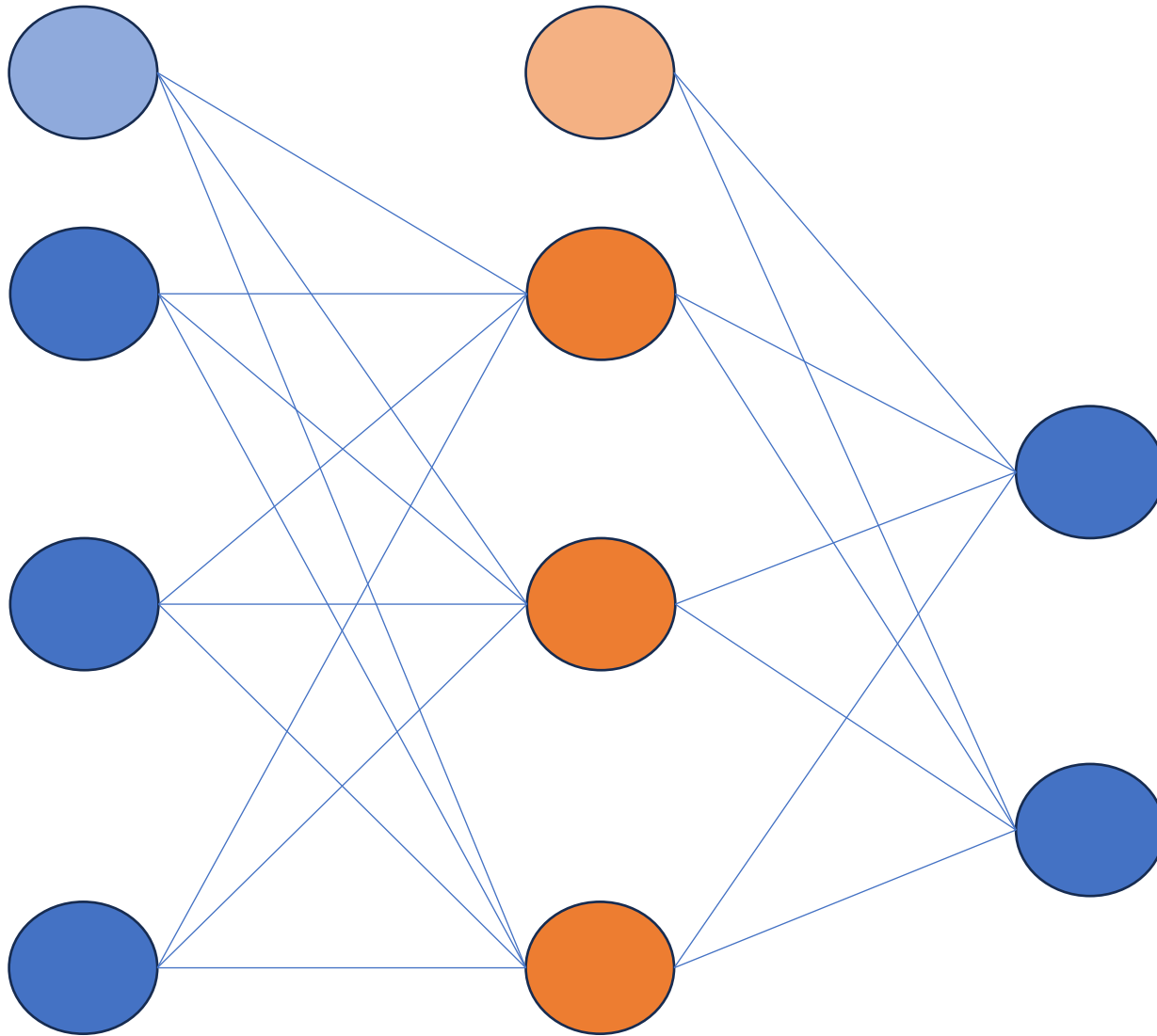
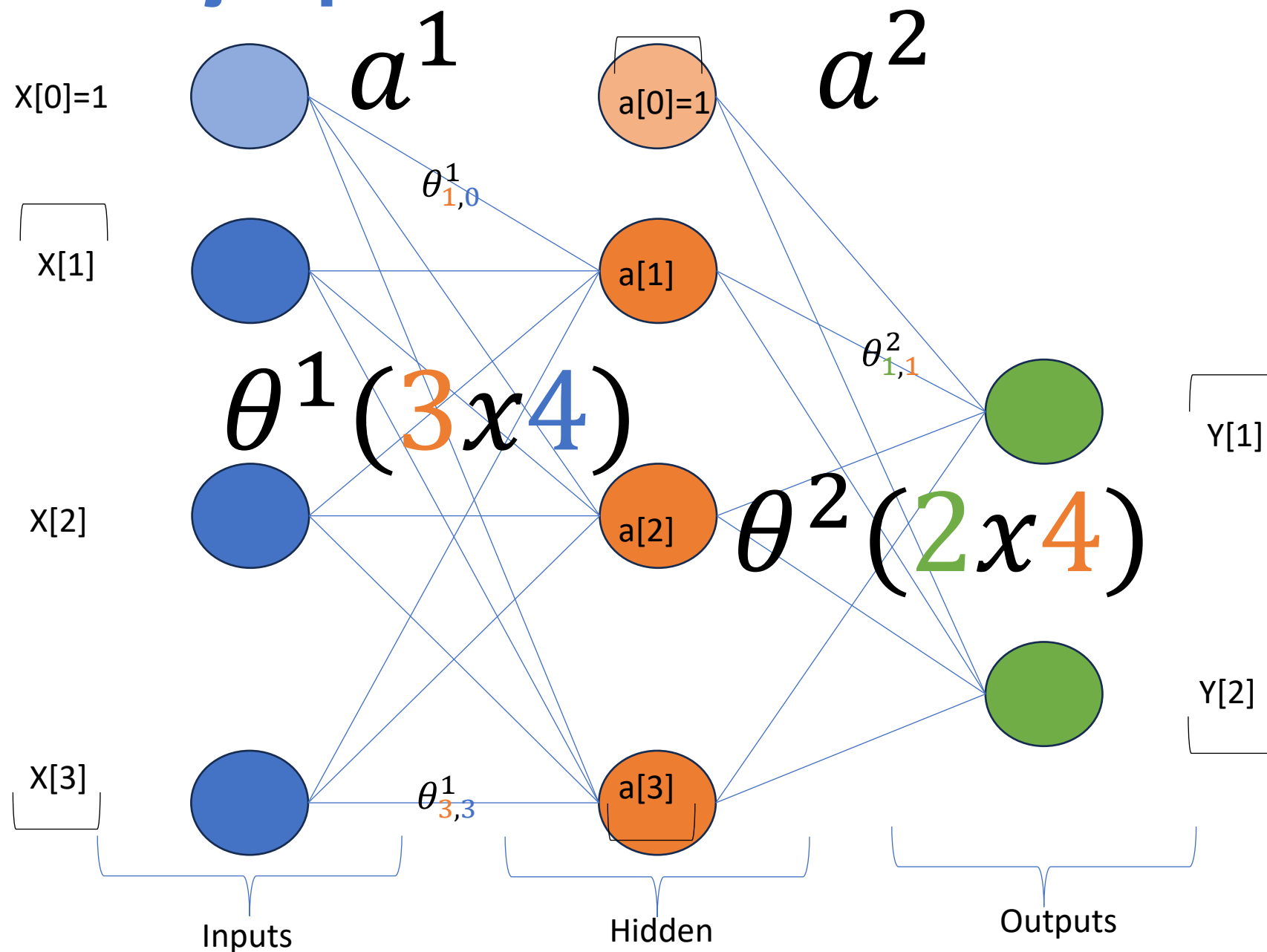


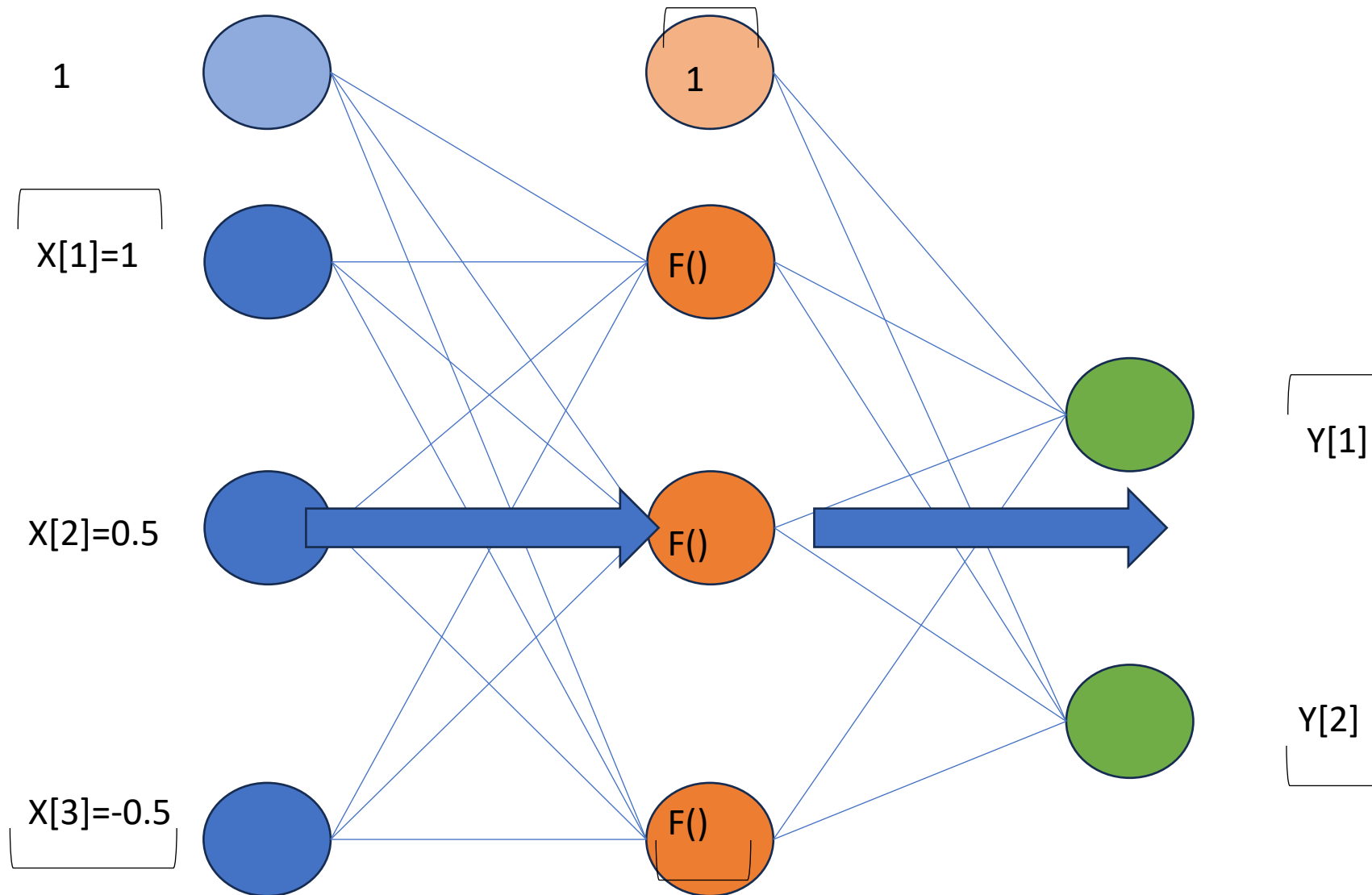
Ejemplo de Red de Neuronas 3x3x2



Ejemplo de Red de Neuronas 3x3x2



Feedforward or Forward Propagation



Matrices Theta y su nomenclatura

$$\theta^1 (3 \times 4)$$

θ				
1	0.1	0.2	0.3	0.4
2	0.5	0.6	0.7	0.8
3	0.9	1	1.1	1.2
	0	1	2	3

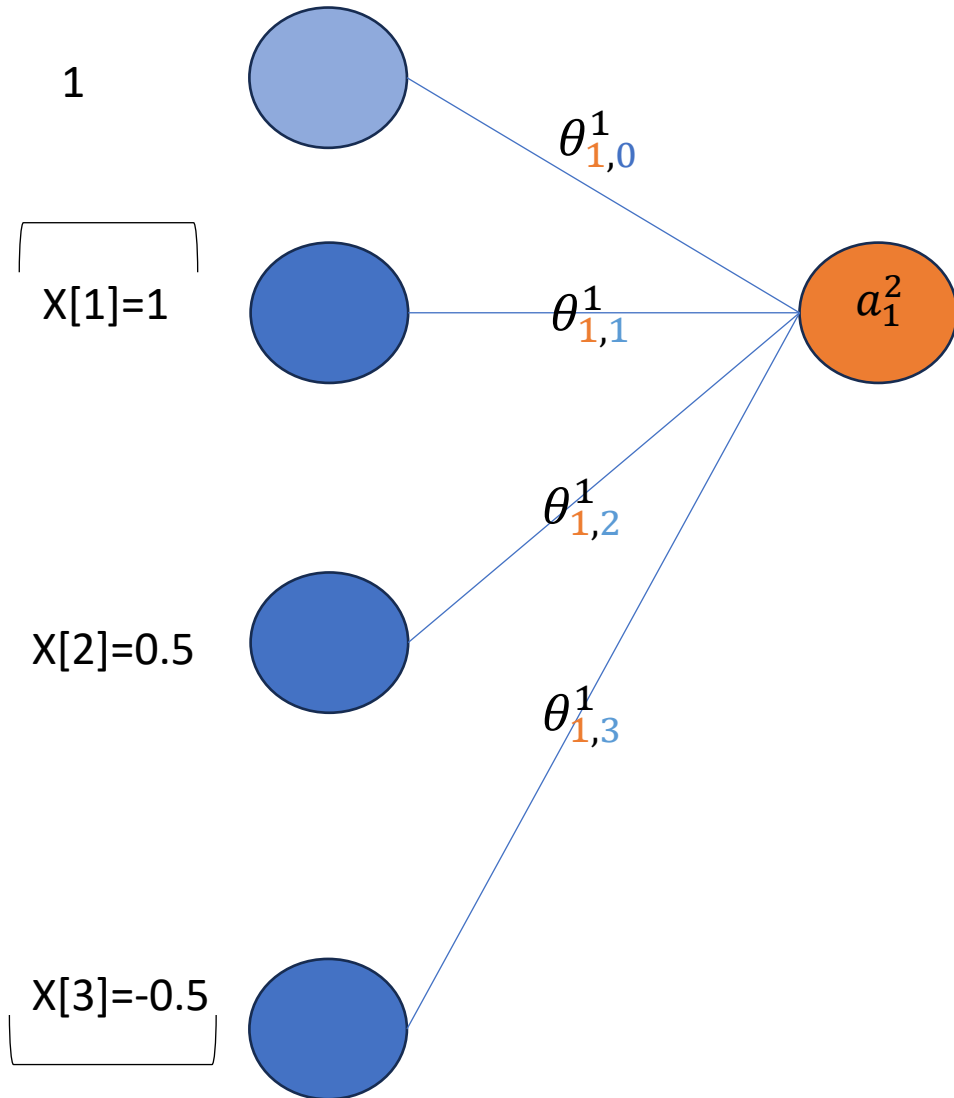
$$\theta^2 (2 \times 4)$$

θ				
1	0.1	0.2	0.3	0.4
2	0.5	0.6	0.7	0.8
	0	1	2	3

The colum 0 (red) is bias (sesgo o el Vector de b integrado en las matrices)

Note: row 0 not exist (in Array implementation with Python, row 1 will be row 0)

Activación de la $a(1)$ en la capa 2

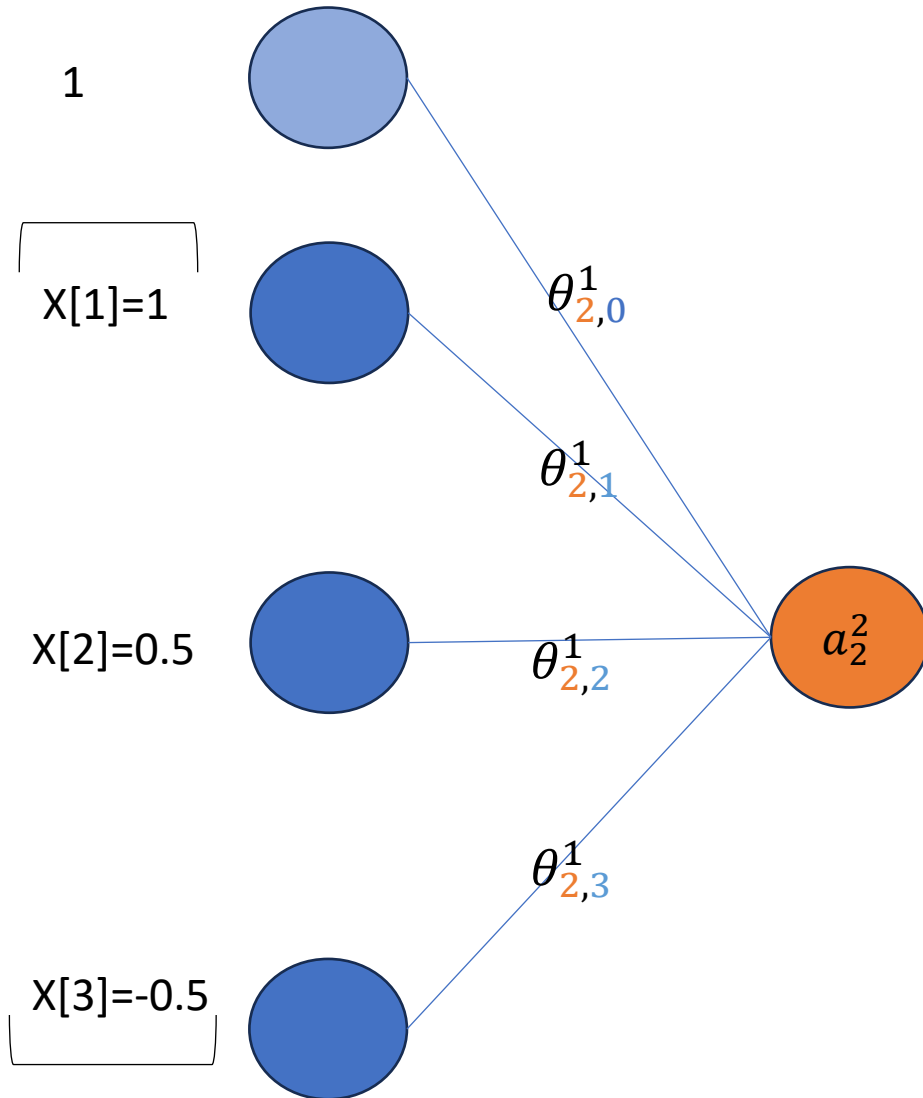


$$\theta^1(3 \times 4)$$

0.1	0.2	0.3	0.4
0.5	0.6	0.7	0.8
0.9	1	1.1	1.2

$$a_1^2 = f\left(\sum_i x_i * \theta_{1,i}^1\right) = f(1 * 0.1 + 1 * 0.2 + 0.5 * 0.3 - 0.5 * 0.4)$$

Activación de la $a(2)$ en la capa 2

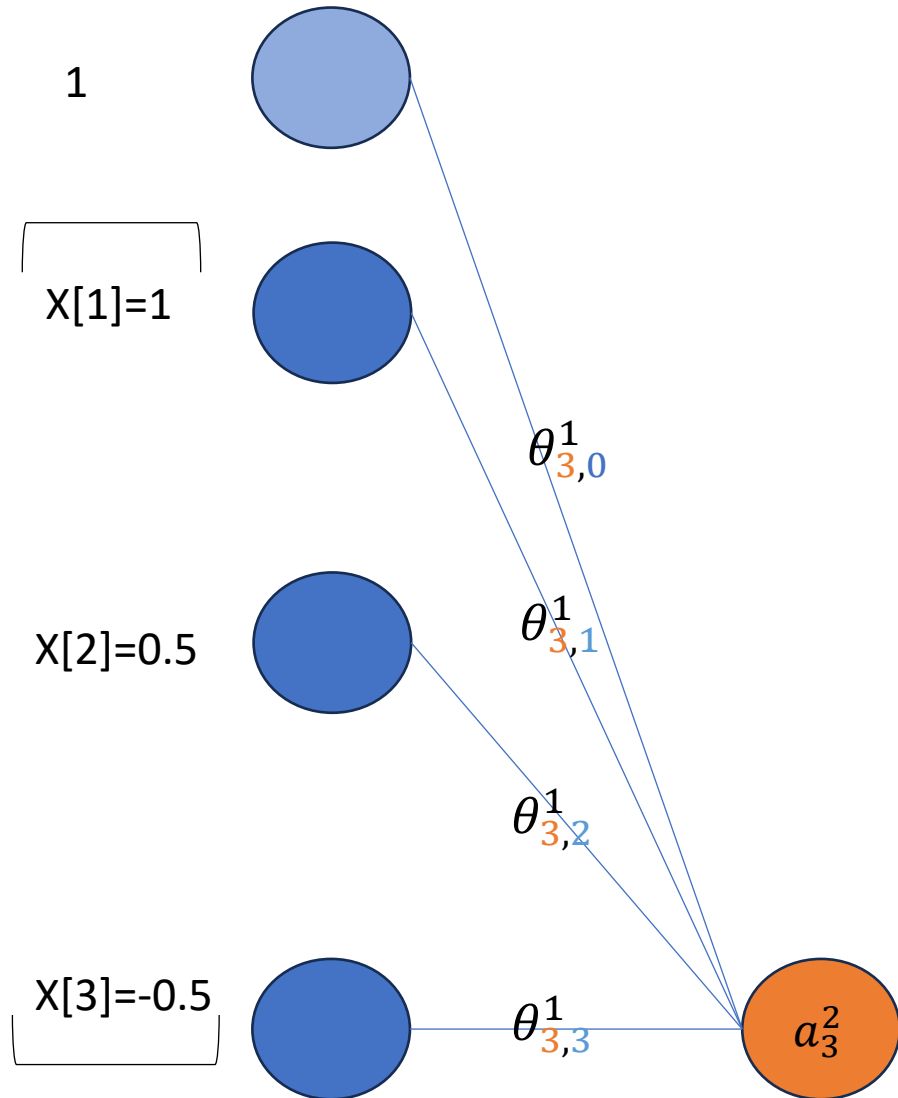


$$\theta^1(3 \times 4)$$

0.1	0.2	0.3	0.4
0.5	0.6	0.7	0.8
0.9	1	1.1	1.2

$$a_2^2 = f\left(\sum_i x_i * \theta_{2,i}^1\right) = f(1 * 0.5 + 1 * 0.6 + 0.5 * 0.7 - 0.5 * 0.8)$$

Activación de la $a(3)$ en la capa 2

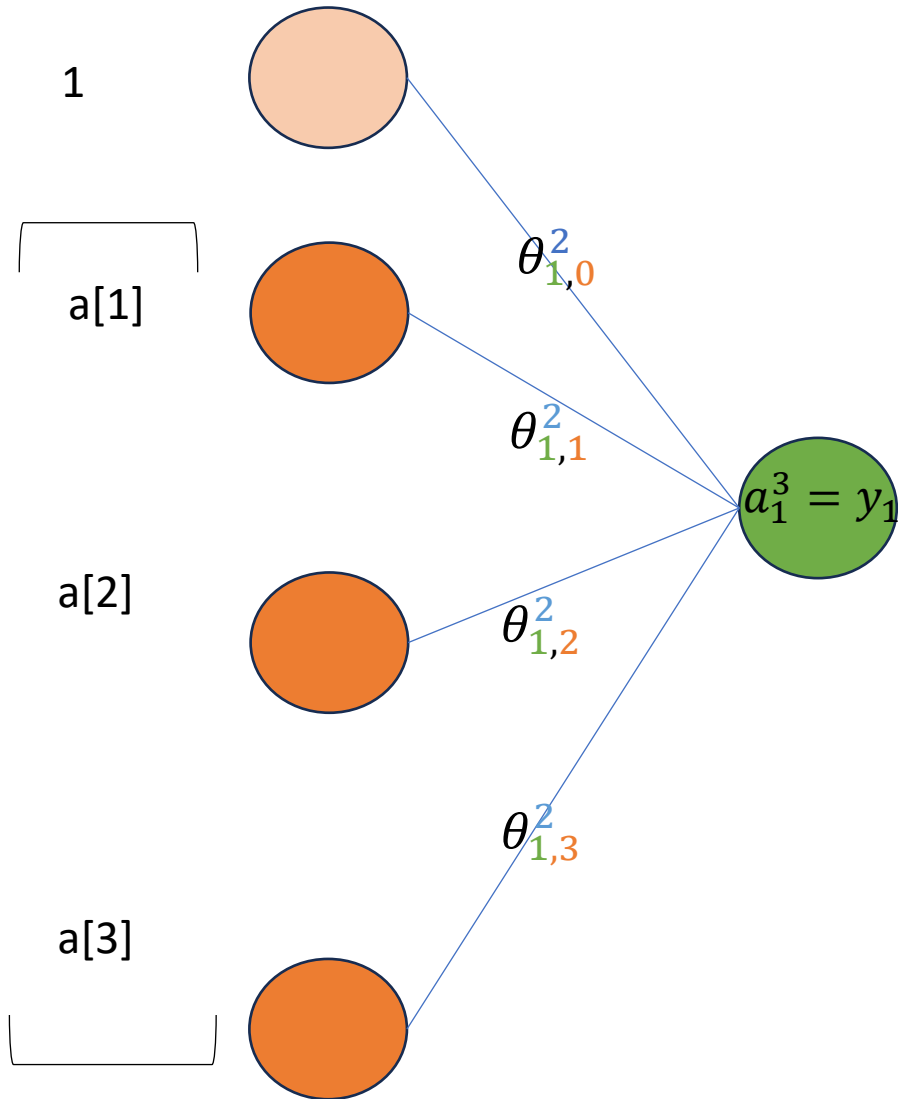


$$\theta^1(3 \times 4)$$

0.1	0.2	0.3	0.4
0.5	0.6	0.7	0.8
0.9	1	1.1	1.2

$$a_3^2 = f\left(\sum_i x_i * \theta_{3,i}^1\right) = f(1 * 0.9 + 1 * 1 + 0.5 * 1.1 - 0.5 * 1.2)$$

Activación de la salida $y(1)$ en capa 3

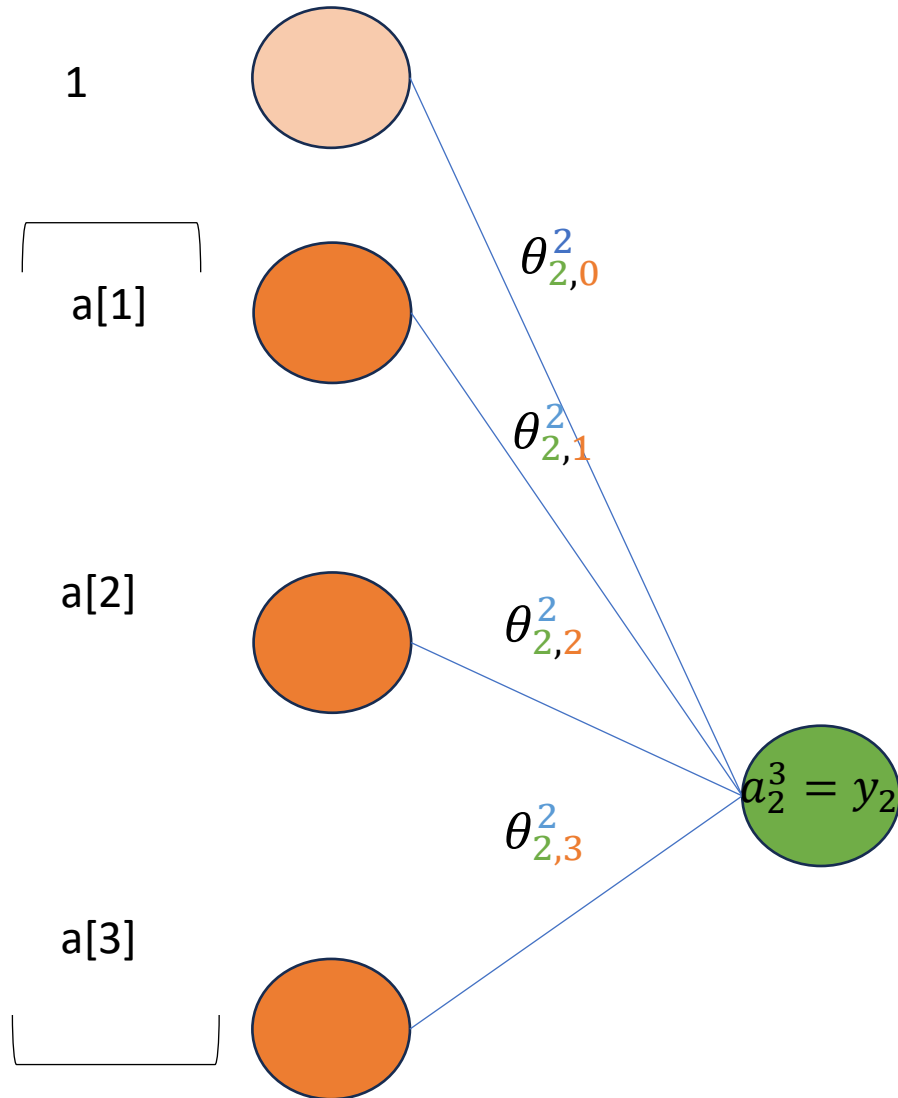


$$\theta^2(2 \times 4)$$

0.1	0.2	0.3	0.4
0.5	0.6	0.7	0.8

$$a_1^3 = y_1 = f\left(\sum_i a_i^2 * \theta_{1,i}^2\right)$$

Activación de la salida $y(2)$ en capa 3



$$\theta^2(2 \times 4)$$

0.1	0.2	0.3	0.4
0.5	0.6	0.7	0.8

$$a_2^3 = y_2 = f\left(\sum_i a_i^2 * \theta_{2,i}^2\right)$$

Implementación iterativa en Python

x

x0=1
x1
x2
x3

a

a0=1
a1
a2
a3

y

y1
y2

For i=0 to NumExamples

 x = [1] U X[i], a = [1] U Zeros(3)

 for j = 1 to len(a) //preservamos a[0]

 for k = 0 to len(x)

 a[j]+=X[k]*Theta1[j,k]

 a[j] = f(a[j])

 for i = 0 to len(y)...

Theta1(3x4)

0	0.1	0.2	0.3	0.4
1	0.5	0.6	0.7	0.8
2	0.9	1	1.1	1.2
	0	1	2	3

Tetha2(2x4)

0	0.1	0.2	0.3	0.4
1	0.5	0.6	0.7	0.8
	0	1	2	3